

Owen Bates

Computer Engineering Student

Westfield, NY
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Education

Rochester Institute of Technology

Bachelor of Science Degree - Computer Engineering – 4th Year - GPA: 3.8

Skills

Technical Abilities/Software: ROS2, NAV2, Linux, Vivado, LTSPICE, Microsoft Office, GameMaker

Programming Languages: C, C++, Python, Java, VHDL ARM Assembly, MIPS Assembly, Verilog, MAKE, CMAKE, MATLAB, VBA, ABB Robotics, GML Language

Work Experience

- **Computer Engineering Intern – Cornell University CAIR Lab (June 2023 – December 2023)**
 - Upgraded the control system for an agricultural robot, improving responsiveness and reliability.
 - Utilized Python, C++, and ROS2, enabling advanced manoeuvrability in agricultural environments.
 - Developed a ROS2-based swarm communication system, enabling seamless interaction and coordination between multiple agricultural robots.
- **Computer Engineering Intern – Cornell University CAIR Lab (May 2024 – August 2024)**
 - Designed a navigation stack for an agricultural robot to autonomously navigate complex agricultural landscapes.
 - Leveraged Python, C++, ROS2, and NAV2 to implement path planning, obstacle avoidance, and localization for reliable autonomous navigation.

Projects

- **Multi-disciplinary Senior Design - Autonomous Transport Sensor System for Visually Impaired Users (January 2026 – Present)**
 - Developing a sensor array and software for obstacle avoidance for transportation of Visually Impaired
 - Gaining experience in cross-functional teamwork, technical communication, and open-source development.

Relevant Coursework

- **Digital Systems Design II**
 - Explored contemporary methods for designing digital systems, including VHDL and Verilog, simulation, synthesis, verification, FPGA implementation, and hands-on projects covering the entire design flow of a MIPS Processor.
- **Applied in C**
 - Explored the C language to implement algorithms used in the numerical solution of common problems encountered in engineering such as computer number representation and roundoff error, algorithms for finding roots of nonlinear equations, interpolation, and data fitting solutions to systems of linear equations.
- **Interface and Digital Electronics (Present)**
 - Explored integrating digital and analog systems through signal conditioning techniques, including amplifiers, analog-to-digital and digital-to-analog conversion methods, and utilized communication protocols such as UART, SPI, and I2C for efficient system interfacing.